

CSC10006 – Introduction to Database

Aggregate Functions and Grouping

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1 Objectives

After completing this lab, students will be able to:

- Aggregate queries using COUNT, SUM, AVG, MIN, MAX
- GROUP BY grouping queries
- GROUP BY and HAVING grouping queries

2 Detailed Instructions

2.1 Using Aggregate Functions in Queries

Aggregate functions: are functions used in queries whose returned results are obtained by combining multiple values together.

Aggregate functions: SQL supports the following aggregate functions:

- **COUNT**: counts the number of rows or the number of values of an attribute.
- **AVG**: calculates the average value.
- **MAX**: returns the maximum value.
- **MIN**: returns the minimum value.
- **SUM**: calculates the total sum.

Note: Except for **COUNT(*)**, which is used to return the number of rows, the other aggregate functions are calculated on a specific attribute across all rows in the query result.

Example 1: Find the total number of teachers in the entire school.

```
SELECT COUNT(*)
```

```
FROM GIAOVIEN
```

Explanation: The COUNT function is used to count the total number of rows in the GIAOVIEN table. Since each row represents one teacher, the result is the total number of teachers in the school.

Example 2: Find the total number of lecturers belonging to the Information Systems department.

```
SELECT COUNT(MAGV)
FROM GIAOVIEN
WHERE MABM = 'HTTT'
```

Example 3: Calculate the number of lecturers who have an academic (professional) supervisor.

```
SELECT COUNT(GVQLCM)
FROM GIAOVIEN
Or
SELECT COUNT(*)
FROM GIAOVIEN
WHERE GVQLCM IS NOT NULL
```

Example 4: Calculate the number of lecturers who act as academic supervisors for other lecturers and belong to the Information Systems (IS) department.

```
SELECT COUNT(DISTINCT GVQLCM)
FROM GIAOVIEN
WHERE MABM = 'HTTT'
```

Note: Use the DISTINCT keyword to count unique records, as demonstrated above.

Example 5: Calculate the average salary of lecturers in the Information Systems department.

```
SELECT AVG(LUONG)
FROM GIAOVIEN GV, BOMON BM
WHERE GV.MABM = BM.MABM AND BM.TENBM = N'Hệ thống thông tin'
```

2.2 Group Queries with GROUP BY:

GROUP BY is used when there is a need to group data in order to perform aggregate calculations. Therefore, GROUP BY is often used together with aggregate functions.

When GROUP BY is used with aggregate functions, the aggregate functions are applied only to rows within the same group.

The groups created using the GROUP BY clause consist of rows that share the same value in the grouping attribute.

Example 6: For each department, display the department code (MAMB) and the number of lecturers in that department..

```
SELECT MABM, COUNT(*)  
FROM GIAOVIEN  
GROUP BY MABM
```

Explanation: The above query operates on the GIAOVIEN table. The GROUP BY MABM clause specifies that the query groups data based on the MABM attribute. It performs the following steps:

- (1) Group the rows of the GIAOVIEN table into separate groups according to the MABM attribute. (Note: all rows within the same group have the same value of the MABM attribute.)
- (2) For each group, output the MABM value and the number of rows in that group.

Example 7: For each lecturer, display the lecturer ID (MAGV) and the number of tasks in which the lecturer has participated.

```
SELECT MAGV, COUNT(*) AS SLCV  
FROM THAMGIADT  
GROUP BY MAGV
```

Example 8: For each lecturer, display the lecturer ID (MAGV) and the number of projects in which the lecturer has participated.

```
SELECT MAGV, COUNT( DISTINCT MADT ) AS SLDT
```

```
FROM THAMGIADT  
GROUP BY MAGV
```

Note: Be careful when using COUNT(...) and COUNT(DISTINCT ...), as illustrated in Examples 7 and 8.

Example 9: For each department, display the number of projects led by lecturers of that department.

```
SELECT GV.MABM, COUNT(*)  
FROM DETAI DT, GIAOVIEN GV  
WHERE DT.GVCNDT = GV.MAGV  
GROUP BY GV.MABM
```

Example 10: For each department, display the department name and the number of lecturers in that department.

```
SELECT BM.TENBM, COUNT(*)  
FROM GIAOVIEN GV, BOMON BM  
WHERE GV.MABM=BM.MABM  
GROUP BY BM.MABM, BM.TENBM
```

Note: All attributes appearing in the SELECT clause must be group attributes. This means that they must satisfy one of the following conditions:

- They are attributes included in the GROUP BY clause, or
- They are derived from aggregate functions.

2.3 Queries with GROUP BY and HAVING

The HAVING clause is used when a query requires conditions that involve the results of grouping, such as the results of aggregate functions.

The condition specified in the HAVING clause is applied after the grouping operation. In this condition, only group attributes may be used.

Example 11: List the departments that have at least two lecturers.

```
SELECT MABM
FROM GIAOVIEN
GROUP BY MABM
HAVING COUNT (*) > 2
```

Example 12: Display the names of lecturers and the number of projects for those lecturers who have participated in three or more projects.

```
SELECT GV.HOTEN, COUNT( DISTINCT TG.MADT) AS SOLUONGDT
FROM GIAOVIEN GV, THAMGIADT TG
WHERE GV.MAGV = TG.MAGV
GROUP BY GV.MAGV, GV.HOTEN
HAVING COUNT( DISTINCT TG.MADT) >= 3
```

2.4 GROUP BY with Extended Attributes

In addition to using existing table attributes, the GROUP BY clause syntax also allows the use of extended attributes, which are formed by applying functions to attributes.

Example 13: Display the number of projects carried out in each year.

```
SELECT YEAR(NGAYBD), COUNT(*)
FROM DETAI
GROUP BY YEAR(NGAYBD)
```

3 In-class exercises

Requirement: Write SQL queries **Q27 → Q34** for the “**Quản lý đề tài**” exercise

Duration: 1 hour.

Q27. Display the number of lecturers and their total salary.

Q28. Display the number of lecturers and the average salary for each department.

Q29. Display the topic name and the number of projects belonging to that topic.

Q30. Display the lecturer name and the number of projects in which the lecturer has participated.

Q31. Display the lecturer name and the number of projects for which the lecturer serves as the project leader.

Q32. For each lecturer, display the lecturer name and the number of relatives of that lecturer.

Q33. Display the names of lecturers who have participated in three or more projects.

Q34. Display the number of lecturers who have participated in the project “*Ứng dụng hoá học xanh*”.